Polynomials & Factoring Error Analysis

TEACHER KEY

Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

1. Subtracting polynomials

ORIGINAL PROBLEM

$(5x^2 + 3x) - (2x^2 - 4x)$

INCORRECT WORKED SOLUTION

$(5x^2 + 3x) - (2x^2 - 4x)$

$= 5x^2 + 3x - 2x^2 - 4x$

$= 3x^2 - x$

A. EXPLAIN THE MISTAKE (1 POINT)

The subtraction must be distributed to BOTH terms: $-(2x^2 - 4x) = -2x^2 + 4x$. The sign of $-4x$ was not changed.

B. SHOW CORRECT WORK (1 POINT)

$(5x^2 + 3x) - (2x^2 - 4x)$

$= 5x^2 + 3x - 2x^2 + 4x$

$= 3x^2 + 7x$

2. Squaring a binomial

ORIGINAL PROBLEM

Expand $(x + 3)^2$

INCORRECT WORKED SOLUTION

$(x + 3)^2$

$= x^2 + 3^2$

$= x^2 + 9$

A. EXPLAIN THE MISTAKE (1 POINT)

$(x + 3)^2$ means $(x + 3)(x + 3)$. The middle terms were skipped: $x^2 + 3x + 3x + 9 = x^2 + 6x + 9$.

B. SHOW CORRECT WORK (1 POINT)

$(x + 3)^2$

$= (x + 3)(x + 3)$

$= x^2 + 3x + 3x + 9$

$= x^2 + 6x + 9$

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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3. Multiplying binomials

ORIGINAL PROBLEM

Expand $(2x - 5)(x + 4)$

INCORRECT WORKED SOLUTION

$(2x - 5)(x + 4)$

$= 2x^2 + 8x - 5x + 20$

$= 2x^2 + 3x + 20$

A. EXPLAIN THE MISTAKE (1 POINT)

The last terms are $(-5)(4) = -20$, not $+20$. Keep track of the negative sign.

B. SHOW CORRECT WORK (1 POINT)

$(2x - 5)(x + 4)$

$= 2x^2 + 8x - 5x - 20$

$= 2x^2 + 3x - 20$

4. Distributing a monomial

ORIGINAL PROBLEM

Expand $3x(x^2 + 2x - 4)$

INCORRECT WORKED SOLUTION

$3x(x^2 + 2x - 4)$

$= 3x^3 + 6x^2 - 4$

A. EXPLAIN THE MISTAKE (1 POINT)

The 3x must multiply EVERY term. The last term should be $3x \cdot (-4) = -12x$.

B. SHOW CORRECT WORK (1 POINT)

$3x(x^2 + 2x - 4)$

$= 3x^3 + 6x^2 - 12x$

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5. Factoring out the GCF

ORIGINAL PROBLEM

Factor $6x^2 + 9x$ completely.

INCORRECT WORKED SOLUTION

$6x^2 + 9x$

$= 3(2x^2 + 3x)$

A. EXPLAIN THE MISTAKE (1 POINT)

The GCF is 3x, not just 3. Factoring out only part of the GCF leaves a common factor of x. Factor completely: $3x(2x + 3)$.

B. SHOW CORRECT WORK (1 POINT)

$6x^2 + 9x$

GCF = $3x$

$= 3x(2x + 3)$

6. Factoring a trinomial

ORIGINAL PROBLEM

Factor $x^2 + 2x - 15$

INCORRECT WORKED SOLUTION

Find two numbers that multiply to -15 and add to 2: 5 and -3

$x^2 + 2x - 15 = (x - 5)(x + 3)$

A. EXPLAIN THE MISTAKE (1 POINT)

The numbers were attached to the wrong signs. Check by multiplying: $(x - 5)(x + 3) = x^2 - 2x - 15$. The correct factors are $(x + 5)(x - 3)$.

B. SHOW CORRECT WORK (1 POINT)

Numbers that multiply to -15 and add to 2: 5 and -3

$x^2 + 2x - 15 = (x + 5)(x - 3)$

Check: $x^2 - 3x + 5x - 15$ ✓

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7. Difference of squares

ORIGINAL PROBLEM

Factor $x^2 - 49$

INCORRECT WORKED SOLUTION

$x^2 - 49$

$= (x - 7)^2$

A. EXPLAIN THE MISTAKE (1 POINT)

A perfect square binomial would give $x^2 - 14x + 49$. A difference of squares factors into a sum and a difference: $(x + 7)(x - 7)$.

B. SHOW CORRECT WORK (1 POINT)

$x^2 - 49$

$= x^2 - 7^2$

$= (x + 7)(x - 7)$

8. Factoring when a ≠ 1

ORIGINAL PROBLEM

Factor $2x^2 + 7x + 3$

INCORRECT WORKED SOLUTION

$2x^2 + 7x + 3$

$= (2x + 3)(x + 1)$

A. EXPLAIN THE MISTAKE (1 POINT)

Check by multiplying: $(2x + 3)(x + 1) = 2x^2 + 5x + 3$, but the middle term should be $7x$. The correct factoring is $(2x + 1)(x + 3)$.

B. SHOW CORRECT WORK (1 POINT)

$2x^2 + 7x + 3$

$= (2x + 1)(x + 3)$

Check: $2x^2 + 6x + x + 3 = 2x^2 + 7x + 3$ ✓

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9. Zero product property

ORIGINAL PROBLEM

Solve $(x - 3)(x + 5) = 0$

INCORRECT WORKED SOLUTION

$(x - 3)(x + 5) = 0$

$x - 3 = 0$ or $x + 5 = 0$

$x = -3$ or $x = 5$

A. EXPLAIN THE MISTAKE (1 POINT)

Solve each factor for x: $x - 3 = 0$ gives $x = 3$, and $x + 5 = 0$ gives $x = -5$. The signs were flipped.

B. SHOW CORRECT WORK (1 POINT)

$(x - 3)(x + 5) = 0$

$x - 3 = 0$ or $x + 5 = 0$

$x = 3$ or $x = -5$

10. Setting equal to zero first

ORIGINAL PROBLEM

Solve $x^2 + 4x = 12$

INCORRECT WORKED SOLUTION

$x^2 + 4x = 12$

$x(x + 4) = 12$

$x = 12$ or $x + 4 = 12$

$x = 12$ or $x = 8$

A. EXPLAIN THE MISTAKE (1 POINT)

The zero product property only works when the product equals ZERO. Move 12 first: $x^2 + 4x - 12 = 0$, so $(x + 6)(x - 2) = 0$.

B. SHOW CORRECT WORK (1 POINT)

$x^2 + 4x = 12$

$x^2 + 4x - 12 = 0$

$(x + 6)(x - 2) = 0$

$x = -6$ or $x = 2$

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points